Linear Approximation and L'Hôpital's Rule

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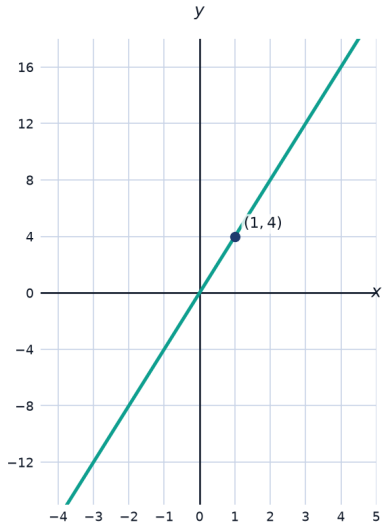
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right) dx$.

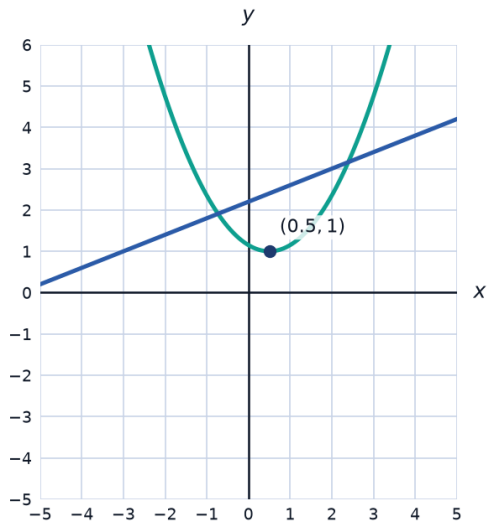
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert the rectangular point $(0,\ -3)$ to polar coordinates.

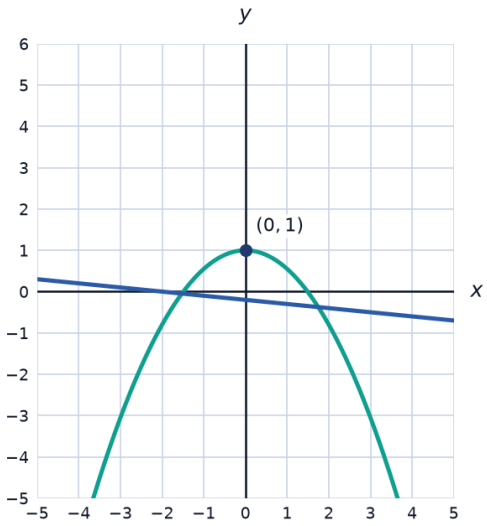
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the speed of a particle with $x = t^2,\ y = t^2$ at $t = 1$.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

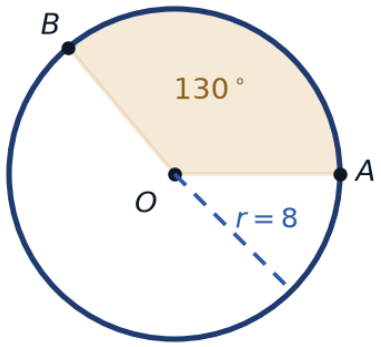
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 2 · MediumEvaluate $\int x^3 \sin(2x)\,dx$ using the Tabular Method.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area Between Curves

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

a. $L(x) = 5 + \frac{1}{10}(x - 25)$

b. $L(x) = 5 + \frac{1}{5}(x - 25)$

c. $L(x) = 5 + \frac{1}{2}(x - 25)$

d. $L(x) = \frac{1}{10}x$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

a. 5.10

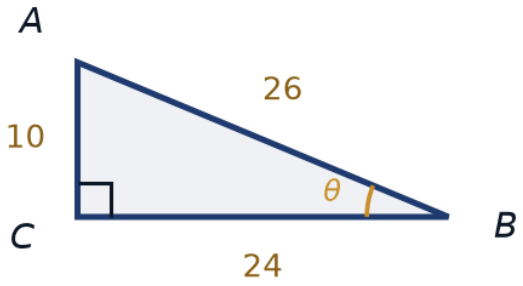
b. 5.05

c. 5.20

d. 5.01

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?



a. An overestimate because $f''(x) < 0$ (concave down)

b. An underestimate because $f''(x) > 0$ (concave up)

c. Exact because f is monotonic

d. Depends on the step size

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x \to c} \frac{f(x)}{g(x)}$?

a. The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$

b. $f(c) = 0$ only

c. $f'(c) = 0$

d. $g(x) > 0$ for all x

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

As $x \to \infty$, which of the following functions grows the fastest?

a. e^x

b. x^{10}

c. $x \ln x$

d. 2^x

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

a. $f(x)$ grows faster than $g(x)$

b. $g(x)$ grows faster than $f(x)$

c. They grow at the same rate

d. Neither function grows

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The midpoint rule approximation for $\int_0^1 \int_0^1 f(x, y)\, dA$ with one rectangle:

a. $f(1, 1)$

b. $f(0, 0)$

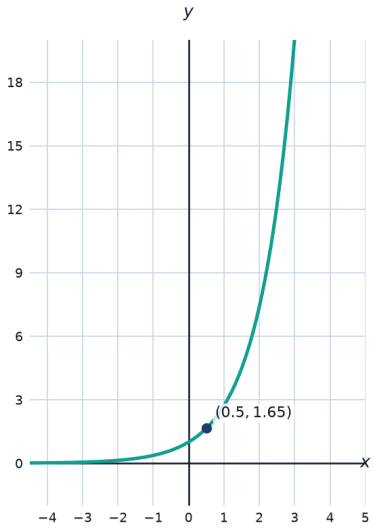
c. $f(1/2, 1/2)$

d. $f(0, 1)$

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For $\frac{dy}{dx} = y,\ y(0) = 1$ (exact solution $y = e^x$), a single Euler step with $h = 0.25$ gives y_1. Compared with the exact value $y(0.25) = e^{0.25}$, Euler's approximation:



a. overestimates the true value

b. equals the true value exactly

c. cannot be compared

d. underestimates the true value

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Using L'Hôpital's rule: $\lim_{x \to 0} \frac{\sin 3x}{5x} =$

a. 3/5

b. 5/3

c. 0

d. 1

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

L'Hôpital advanced: $\lim_{x \to 0^+} x^x =$

a. 1

b. 0

c. e

d. DNE

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use L'Hôpital's rule: $\lim_{x \to 0} \frac{e^x - 1}{\sin x}$ equals:

a. 0

b. 1

c. e

d. undefined

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

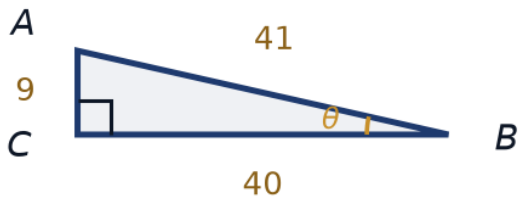
b. 1/0

c. 0/0

d. 1/1

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Best linear approximation to $e^x \sin x$ at $x = 0$:



a. x

b. $1 + x$

c. $2x$

d. $x + x^2$

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

LIATE rule for IBP: first choice for u among x^2, e^x, $\ln x$:

a. e^x

b. x^2

c. $\ln x$

d. any of them

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Linear approximation: $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$. This is accurate near:

a. (a, b) if f is differentiable

b. far from (a, b)

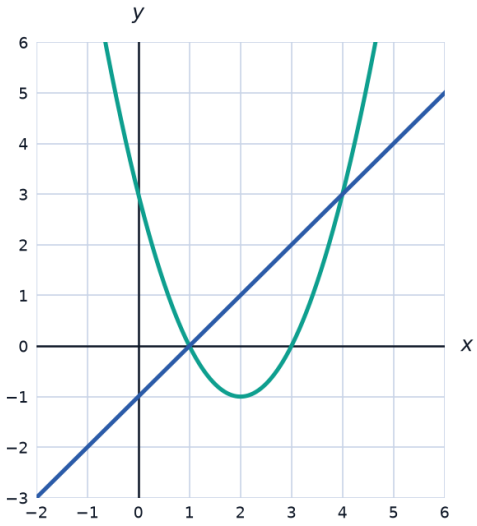
c. only when f is quadratic

d. when $ablaf = 0$

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?



a. 0

b. 1

c. -1

d. 2

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

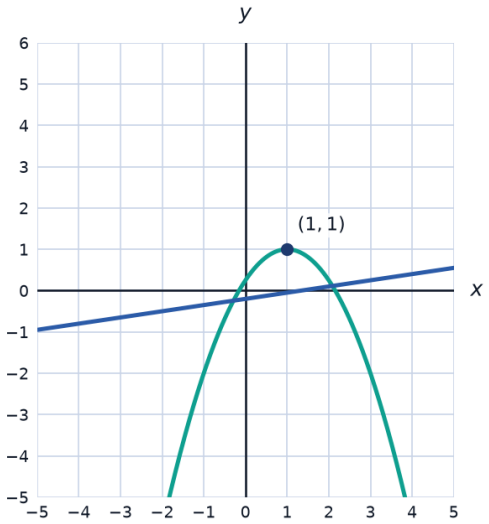
c. $\frac{1}{4}$

d. $\frac{2}{3}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.



a. $-\frac{1}{2}+\frac{e^4}{2}$

b. $-1+e^4$

c. $\frac{1}{2}+\frac{e^4}{2}$

d. $-\frac{1}{4}+\frac{e^4}{4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area Between Curves

Use $x=u+v, y=u-v$ to compute $\iint_D (x-y)e^{x+y}\,dA$ where D is the square $|x-y|\le 1, |x+y|\le 1$:

a. 0

b. $2(e-e^{-1})$

c. $(e-1)^2$

d. $4(e-e^{-1})/e$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$

d. $-\frac{3}{2}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Limits Graphically and Numerically

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\,dV =$

a. 1/6

b. 1/24

c. 1/12

d. 1/8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Definition of the Derivative

Evaluate $\iiint_B (x^2 + y^2 + z^2)\,dV$ where B is the ball $x^2 + y^2 + z^2 \le 1$.

a. $\frac{4\pi}{5}$

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

If $f'(x) = (x - 3)$, on which interval is f increasing?

a. $(3, \infty)$

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

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